# CBL Deep Learning: Image Classification Challenge

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| **PS No. and Title:** PS-41, Kannada Handwritten Digit Recognition | **Date:** 23 October 2026 | |
| **GitHub:** https://github.com/pramodhvijay6-cmd/[add repo name] | **Colab:** https://colab.research.google.com/drive/1aiIGS835mZpKXwpda_Vf1x9p4EBSgd8F?usp=sharing | |

## 1. Problem and Objective

Kannada is written with its own numerals, and forms, records and exam sheets need a reader that works on new handwriting, not only the writers it was trained on. The task is to classify 28 × 28 handwritten Kannada digits (0 to 9) with a CNN. Success is judged by test accuracy and macro F1, reported separately on a standard test set and on Dig-MNIST (different writers), and by the size of the drop between the two, which measures domain shift.

## 2. Dataset and Preprocessing

Kannada-MNIST (Prabhu, 2019), loaded from the authors' GitHub repository (vinayprabhu/Kannada_MNIST), MNIST-format .npy files. Train: 60,000 images (6,000 per class). Standard test: 10,000 labelled images (1,000 per class). Dig-MNIST: 10,240 labelled images (1,024 per class) written by other volunteers, with visibly bolder strokes and different proportions, used only as the domain-shift test set. A stratified 90/10 split of the training set (seed 42) gave 54,000 train and 6,000 validation images. Pixels were scaled to 0 to 1. Classes are perfectly balanced, so no imbalance handling was needed. The Augmented model used random rotation (factor 0.08, about ±29°), zoom (0.15) and translation (10%) during training only. Flips were not used because they change the shape of a Kannada digit. The test sets were never augmented or used for tuning.

## 3. Methodology

Both models share one CNN trained from scratch: two blocks of two 3 × 3 convolutions (32 then 64 filters) with batch normalisation, max pooling and dropout (0.25), then a 128-unit dense layer with batch normalisation, dropout (0.4) and a 10-way softmax (468,714 parameters; 0.23 ms per image, batched, on a T4 GPU). The Baseline has no augmentation. The Augmented model puts the augmentation layers in front of the same CNN. The model with the higher validation accuracy is the main model, chosen before looking at either test set.

| Setting | Value | Setting | Value |
|---|---|---|---|
| Loss | Sparse categorical cross-entropy | Epochs (max) | 5 Baseline, 6 Augmented |
| Optimiser | Adam, lr 1e-3 | Early stopping | val_loss, patience 2, restore best |
| Batch size | 256 | LR schedule | ReduceLROnPlateau, ×0.5, patience 1 |
| Seed | 42 | Environment | Python 3.13, TF 2.21.0, T4 GPU |

*Table 1. Hyperparameters.*

## 4. Results

| Model | Val acc | Test acc | Test macro F1 | Dig acc | Dig macro F1 | Train time |
|---|---|---|---|---|---|---|
| Baseline (main) | 99.53 | 97.85 | 0.98 | 77.55 | 0.77 | 0.72 min |
| Augmented | 98.07 | 94.34 | 0.94 | 81.39 | 0.82 | 0.76 min |

*Table 2. Accuracy in % on each split (Val = 6,000 validation images; Test = 10,000; Dig = Dig-MNIST, 10,240); macro F1 as a fraction. Training times on a T4 GPU.*

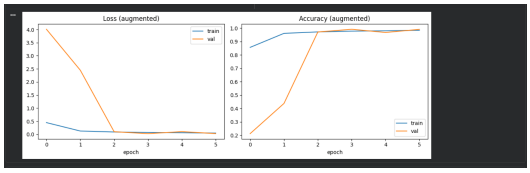



*Fig. 1. Learning curves (Augmented model, first run).*

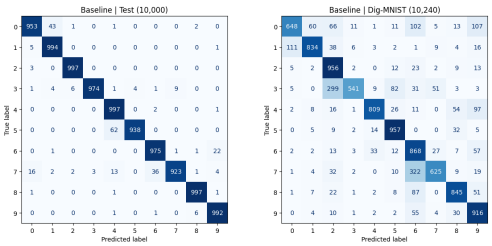



*Fig. 2. Baseline confusion matrices, test (left) and Dig-MNIST (right), first run.*

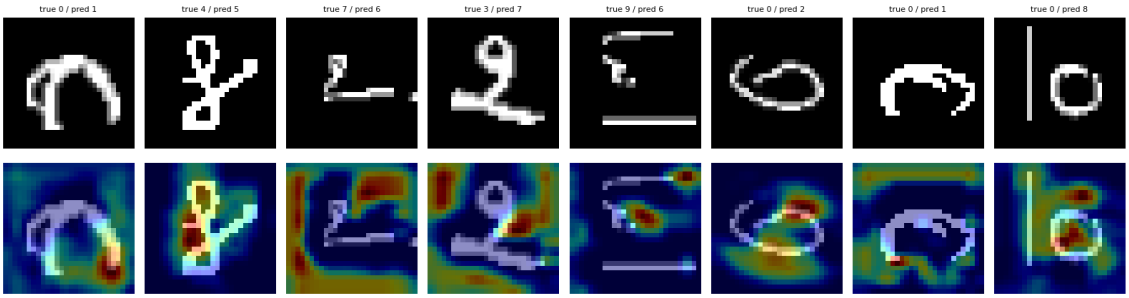



*Fig. 3. Grad-CAM on eight random Dig-MNIST errors (Augmented model, first run, seed 42).*

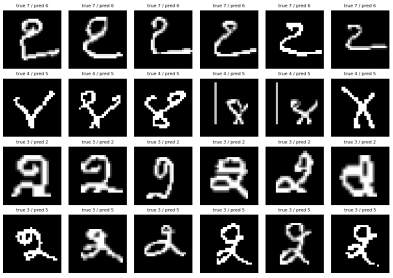


*Fig. 4. Misclassified Dig-MNIST images (Augmented model, first run). Rows: true 7 predicted 6, true 4 predicted 5, true 3 predicted 2, true 3 predicted 5. Figs. 1 to 4 come from the first training run; GPU training is not fully repeatable, so its counts differ slightly from Table 2 (Baseline there: 97.85% test, 77.55% Dig-MNIST; first run: 97.40% and 78.12%).*

## 5. Analysis and Discussion

Both models fit the validation set well, but validation accuracy in epoch 1 was low (0.53 Baseline, 0.10 Augmented) while training accuracy was already high; this is the batch-normalisation statistics settling, and it stabilised from epoch 3. The Augmented model was still improving at its 6-epoch cap (validation loss 0.056), so its validation score is a lower bound. The specific challenge is domain shift: accuracy fell from 97.85% on the test set to 77.55% on Dig-MNIST for the Baseline (a drop of 20.30 points) and from 94.34% to 81.39% for the Augmented model (12.95 points). Augmentation therefore traded 3.51 points of test accuracy for 3.84 points on the shifted writers, but it was capped at 6 epochs, so this trade-off may shrink with longer training. Dig-MNIST strokes are bolder and proportions differ, so features tied to the training writers' style transfer poorly. The most confused pairs on Dig-MNIST were 7 predicted as 6 (322 images), 3 as 2 (299), 1 as 0 (111), 0 as 9 (107) and 0 as 6 (102) for the Baseline (Fig. 2). In Figs. 3 and 4 many errors are faint, broken or stray-stroke drawings, and in several of the eight Grad-CAM maps the heat sits on empty background or edges rather than on the strokes, a small sample that suggests weak cues for unfamiliar writing styles.

## 6. Real-world Relevance, Limitations and Ethics

The model could support Kannada form digitisation, exam-sheet reading and education apps. It handles only isolated digits on clean 28 × 28 images, was trained on one dataset, and drops sharply on unseen writing styles, so it should not be used where a wrong digit has a cost (amounts, IDs) without human checking. Handwriting samples can identify people, so the collection of any new data needs consent.

## 7. Conclusion and Future Work

The Baseline CNN reached 97.85% on the test set but only 77.55% on Dig-MNIST, a 20.30-point domain-shift gap that augmentation narrowed to 12.95 points. Next I would train both models for 12 epochs and add stroke-thickness (dilation) and elastic-distortion augmentation to imitate Dig-MNIST writers.

Data: V. U. Prabhu, “Kannada-MNIST: A new handwritten digits dataset for the Kannada language,” arXiv:1908.01242, 2019. AI use declaration: Claude (Anthropic) was used for guidance and code drafting; all runs and results were produced and checked by the student.